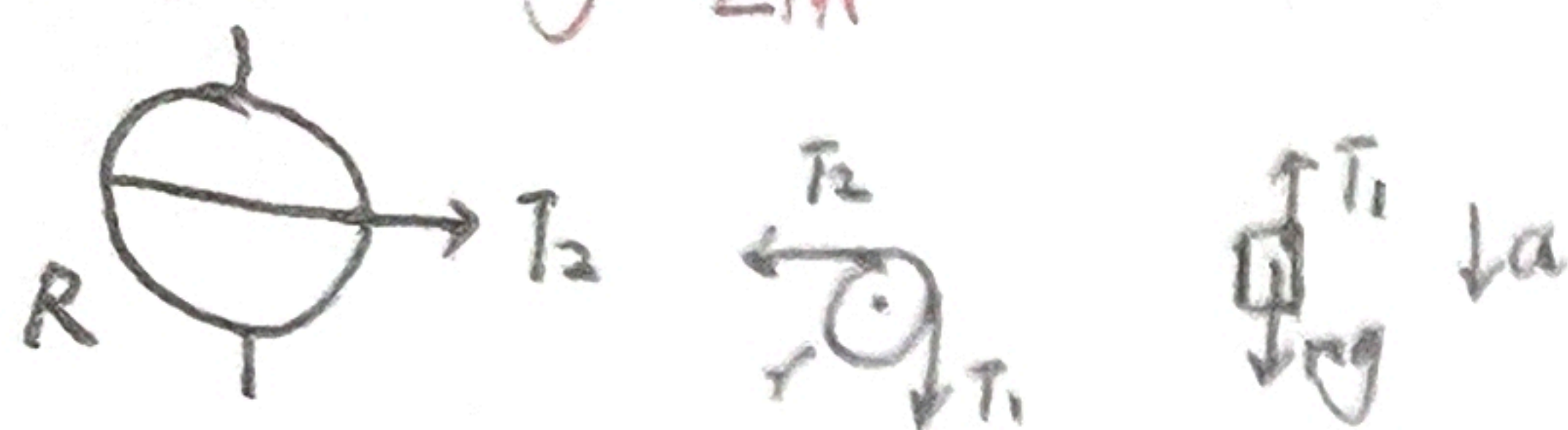


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$$1) \begin{cases} mg - T_1 = ma \\ T_1 r - T_2 r = J\beta_1 = J\frac{a}{r} \\ T_2 R = (\frac{2}{5}MR^2)\beta_2 = (\frac{2}{5}MR^2)\frac{a}{R} \end{cases}$$

$$(mg - ma)r^2 - \frac{2}{5}Mar^2 = J a R$$

$$a = \frac{mg r^2}{J + \frac{2}{5}MR^2 + mr^2}$$

$$a = \frac{mg r^2}{J + \frac{2}{5}MR^2 + mr^2}$$

$$2) v^2 = 2ah$$

$$v = \sqrt{\frac{2mg r^2 h}{J + \frac{2}{5}MR^2 + mr^2}}$$

二. 1) ~~波函数~~ $y_1(x,t) = A \cos[2\pi(\nu t + \frac{x}{\lambda})]$

$x=0$ 处质点的振动方程 $y(0,t) = A \cos 2\pi \nu t$

$$y_2(x,t) = A \cos[2\pi(\nu t - \frac{x}{\lambda})]$$

$$2) y_{\text{总}} = y_1 + y_2 = A \cos[2\pi(\nu t + \frac{x}{\lambda})] + A \cos[2\pi(\nu t - \frac{x}{\lambda})]$$

$$= 2A \cos(2\pi \nu t) \cos \frac{2\pi x}{\lambda}$$

$$3) \frac{2\pi}{\lambda} \Delta x = k\pi \quad \Delta x = \frac{k\lambda}{2} \quad k=0,1,2,\dots$$

$$\frac{2\pi}{\lambda} \Delta y = (k+\frac{1}{2})\pi \quad \Delta y = (\frac{k}{2} + \frac{1}{4})\lambda \quad k=0,1,2,\dots$$

三. 设电荷线密度为 λ , 由高斯定理

$$\vec{E} = \frac{\lambda \vec{r}}{2\pi\epsilon_0 r^2} \quad r_1 < r < r_2$$

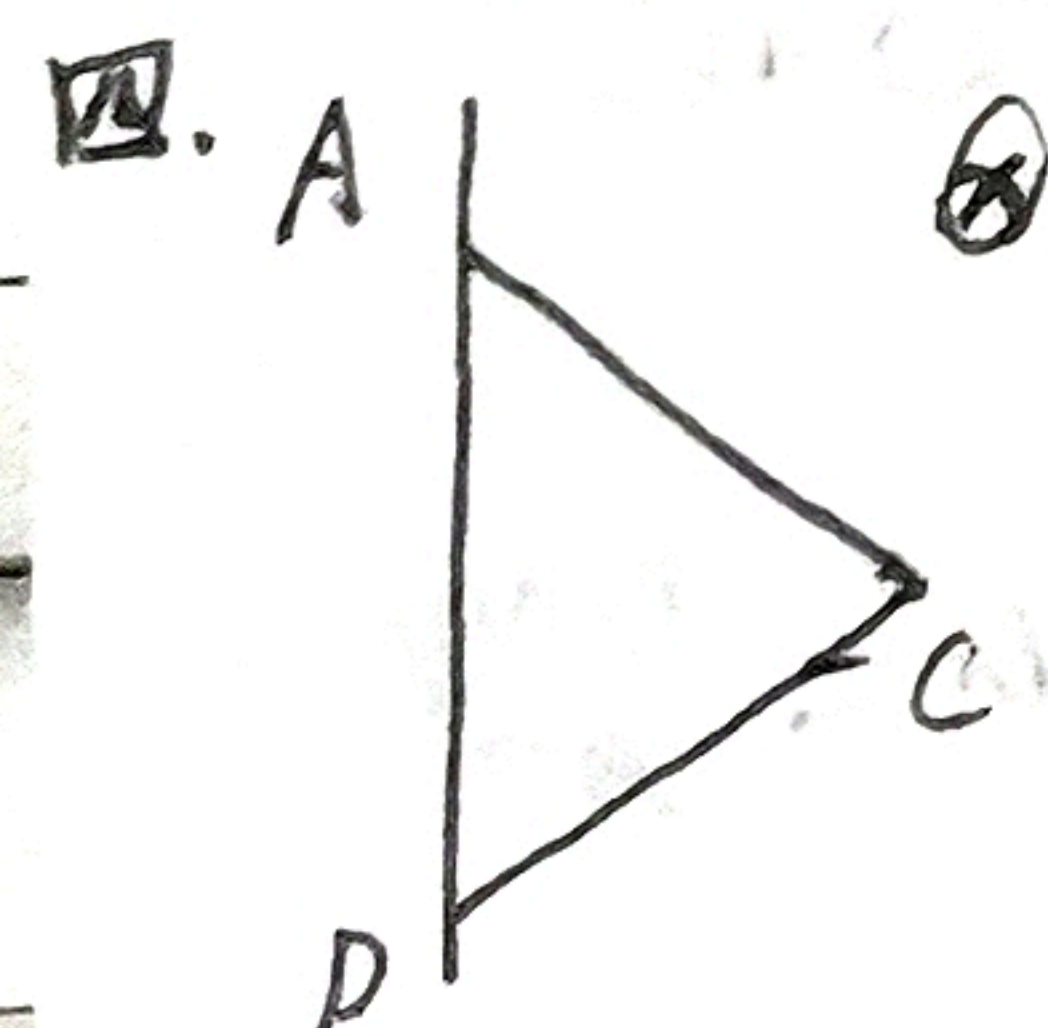
$$\vec{E} = 0 \quad \text{as } r < r_1$$

$$\text{导线表面 } E_1 = \frac{1}{2}(\vec{E}_1 + \vec{E}_2) = \frac{\lambda}{4\pi\epsilon_0 r_1} \quad \text{同轴内表面 } E_2 = -\frac{\lambda}{4\pi\epsilon_0 r_2}$$

$$V = \int_{r_1}^{r_2} E dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_2}{r_1} \quad \lambda = 2\pi\epsilon_0 V \ln \frac{r_2}{r_1} = 2\pi \times 8.85 \times 10^{-12} \times 850 \times \ln \frac{0.134 \times 10^{-9}}{0.02}$$

$$E_1 = \frac{2\pi\epsilon_0 V \ln \frac{r_2}{r_1}}{2\pi\epsilon_0 r_1} = \frac{V}{r_1} \ln \frac{r_2}{r_1} < 0 \quad \text{算出来是负的?!?}$$

$$E_2 = \frac{V}{r_2} \ln \frac{r_2}{r_1}$$



$$d\vec{L} = (\vec{r} \times \vec{B}) \cdot d\vec{L}$$

$$1) \Phi(t) = \vec{B} \cdot \vec{S} = BS \cos \omega t \quad S = \frac{\sqrt{3}}{4} a^2$$

$$\mathcal{E} = -\frac{d\Phi(t)}{dt} = \frac{\sqrt{3}}{4} Ba^2 \omega \sin \omega t$$

$$\mathcal{E}_{AC} = \mathcal{E}_{CD} = \frac{1}{2} \mathcal{E} = \frac{\sqrt{3}}{8} Ba^2 \omega \sin \omega t$$

$$2) I = \frac{\mathcal{E}}{R} = \frac{\sqrt{3}}{4R} Ba^2 \omega \sin \omega t$$

$$3) U_{CA} = U_{DC} = \mathcal{E}_{CA} - \frac{1}{3} IR$$

$$= \frac{\sqrt{3}}{24} Ba^2 \omega \sin \omega t$$

$$U_{DA} = \frac{1}{3} IR = \frac{\sqrt{3}}{12} Ba^2 \omega \sin \omega t$$

$$五. f(v)dv = \frac{dN}{Ndv} \cdot dv = A \exp\left(-\frac{mv^2}{2kT}\right) v^2 dv \quad A = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2}$$

$$f(v_x, v_y, v_z) = f(v_x^2 + v_y^2 + v_z^2)$$

$$f(v_x, v_y) = f(v_x^2 + v_y^2) = g(v_x^2)g(v_y^2)$$

$$\ln f(v^2) = \ln g(v_x^2) + \ln g(v_y^2)$$

$$\frac{\partial \ln f(v^2)}{\partial v_x^2} = \frac{\partial \ln g(v_x^2)}{\partial v_x^2} \quad \frac{\partial \ln f(v^2)}{\partial v_x^2} = \frac{\partial \ln f(v^2)}{\partial v^2} \frac{\partial v^2}{\partial v_x^2} = \frac{\partial \ln f(v^2)}{\partial v^2}$$

$$\text{同理} \quad \frac{\partial \ln f(v^2)}{\partial v^2} = \frac{\partial \ln g(v_y^2)}{\partial v_y^2} = \frac{\partial \ln f(v^2)}{\partial v_y^2} = -\alpha$$

$$g(v_i^2) = C e^{-\alpha v_i^2}$$

$$\int C e^{-\alpha v_i^2} dv_i = 1 \quad C \sqrt{\frac{\pi}{\alpha}} = 1 \quad C = \sqrt{\frac{\alpha}{\pi}}$$

$$\overline{v_i^2} = \frac{1}{\sqrt{\frac{\alpha}{\pi}}} \int v_i^2 e^{-\alpha v_i^2} dv_i = \frac{1}{2\alpha} \quad \frac{1}{2} m \overline{v_i^2} = \frac{1}{2} kT$$

$$\alpha = \frac{m}{2kT} \quad C = \sqrt{\frac{m}{2\pi kT}}$$

$$g(v_i^2) = \sqrt{\frac{m}{2\pi kT}} e^{-\frac{m}{2kT} v_i^2}$$

$$f(v_x, v_y) = \frac{m}{2\pi kT} e^{-\frac{m}{2kT} (v_x^2 + v_y^2)}$$

$$f(v) = 2\pi v \frac{m}{2\pi kT} e^{-\frac{m}{2kT} v^2}$$

$$\text{最概然速率 } v_m \quad \frac{df(v)}{dv} = \frac{m}{2\pi kT} \left(2\pi e^{-\frac{m}{2kT} v^2} + 2\pi v \left(-\frac{m}{kT}\right) e^{-\frac{m}{2kT} v^2} \right)$$

$$= \frac{m}{kT} e^{-\frac{m}{2kT} v^2} \left(1 - \frac{m}{kT} v^2 \right) = 0$$

$$v_m = \sqrt{\frac{kT}{m}}$$

$$\text{平均速率 } \bar{v} = \int v f(v) dv = \frac{m}{kT} \int v^2 e^{-\frac{m}{2kT} v^2} dv$$

$$= \frac{m}{kT} \frac{\sqrt{\pi}}{2} \left(\frac{kT}{m}\right)^{3/2}$$

$$\int e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$$

$$-\int x^2 e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}} \left(-\frac{1}{2}\right) \alpha$$

于是

$$= \sqrt{\frac{\pi}{m}} \sqrt{\frac{2\pi kT}{m}}$$

$$\text{平均速率平方 } \overline{v^2} = \int v^2 f(v) dv = \frac{m}{kT} \int v^3 e^{-\frac{m}{2kT} v^2} dv \quad \text{令 } A = \frac{m}{2kT}$$

$$v_{rms} = \sqrt{\frac{2kT}{m}}$$

$$= A \int v^2 e^{-Av^2} dv^2 = A \int_0^\infty x e^{-Ax} dx = -\int x d e^{-Ax}$$

$$= \left(x e^{-Ax} \Big|_0^\infty - \int_0^\infty e^{-Ax} dx \right) = \int_0^\infty e^{-Ax} dx$$

$$= 1 + \frac{2kT}{m}$$

$$\int_0^\infty x^n e^{-Ax} dx = \frac{n!}{A^{n+1}}$$

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二、1) 物理意义: 将分子看作有一定大小的刚性球, 则每个球的自由活动空间不再是 V , 设 1mol 气体有 V_m 体积, 分子自由活动空间为 $V_m - b$
 a 表示 1mol 气体的吸引力修正

$$(P + \frac{a}{V^2})(V - b) = RT \quad P = \frac{RT}{V-b} - \frac{a}{V^2}$$

临界点处 $(\frac{\partial P}{\partial V})_T = 0 \quad (\frac{\partial^2 P}{\partial V^2})_T = 0$

$$(\frac{\partial P}{\partial V})_T = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3} \quad (\frac{\partial^2 P}{\partial V^2})_T = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4}$$

在临界点处

$$P_c = \frac{RT_c}{V_c - b} - \frac{a}{V_c^2}, \quad -\frac{RT_c}{(V_c - b)^2} + \frac{2a}{V_c^3} = 0, \quad \frac{2RT_c}{(V_c - b)^3} - \frac{6a}{V_c^4} = 0$$

解得 $P_c = \frac{a}{27b^2}, \quad V_c = 3b, \quad RT_c = \frac{8a}{27b}$ 即 $\frac{RT_c}{P_c V_c} = \frac{8}{3}$

临界点

设无量纲参量 $\tilde{P} := \frac{P}{P_c}, \quad \tilde{V} := \frac{V}{V_c}, \quad \tilde{T} := \frac{T}{T_c}$ 得到

临界系数!!!

临界指数 α, β, γ 不会

$$(\tilde{P} + \frac{3}{\tilde{V}^2})(3\tilde{V} - 1) = 8\tilde{T}$$

在临界点附近引入 $\pi := \frac{P - P_c}{P_c} = \tilde{P} - 1, \quad \phi := \frac{V - V_c}{V_c} = \tilde{V} - 1, \quad \tau := \frac{T - T_c}{T_c} = \tilde{T} - 1$

代入

$$(1 + \pi + \frac{3}{(1 + \phi)^2})(3(1 + \phi) - 1) = 8(1 + \tau)$$

$$1 + \pi = \frac{8(1 + \tau)}{3(1 + \phi) - 1} - \frac{3}{(1 + \phi)^2} = \frac{4(1 + \tau)}{1 + \frac{3\phi}{2}} - \frac{3}{1 + 2\phi + \phi^2}$$

$$\pi = 4\tau - 6\phi + 9\tau\phi - \frac{3}{2}\phi^3 + \dots$$

$$\pi \approx 4\tau - 6\phi - \frac{1}{2}\phi^3$$

$$\beta = \frac{1}{2}, \quad \gamma = 3, \quad \alpha = 0$$

$$\begin{cases} \frac{1}{f_1} - \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{s_1} \\ \frac{1}{f_1} - \frac{1}{f_2} = \frac{1}{s_2} \\ \frac{1}{f_1} = \frac{1}{s_3} \end{cases} \Rightarrow$$

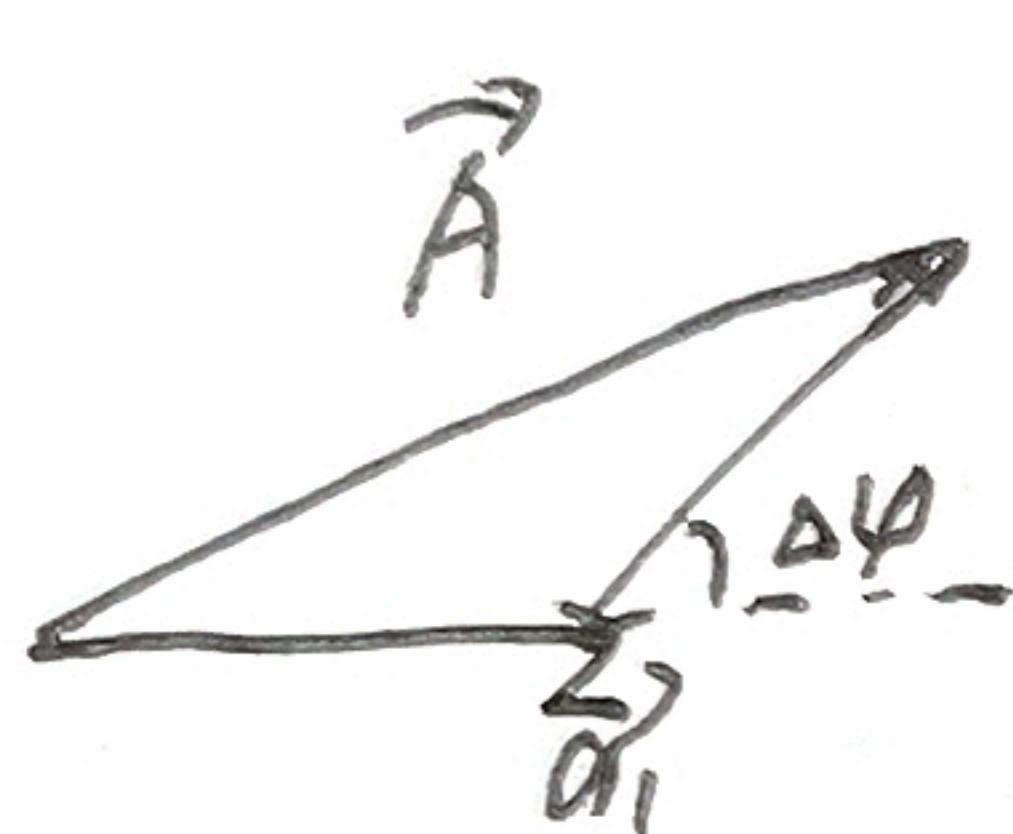
$$\frac{1}{a} - \frac{1}{4a} = \frac{1}{2a}$$

对单缝. 单缝衍射因子为

$$\vec{a}_1(\theta) = U_0 \frac{\sin u}{u} \quad \vec{a}_2(\theta) = 2U_0 \frac{\sin 2u}{2u} \quad u = \frac{\pi a \sin \theta}{\lambda}$$

θ 方向的衍射光 $b = d \sin \theta = 4a \sin \theta$

$$\Delta \phi = 4ka \sin \theta = 4 \frac{2\pi}{\lambda} a \sin \theta = 8u$$



$$\vec{A} = \vec{a}_1 + \vec{a}_2 - 2a_1 a_2 \cos(\pi - \delta \phi)$$

$$= (U_0 \frac{\sin u}{u})^2 + (2U_0 \frac{\sin 2u}{2u})^2 + 4U_0^2 \frac{\sin u}{u} \frac{\sin 2u}{2u} \cos 8u$$

$$= 4U_0^2 \frac{2 \sin u \cos u}{2u} + 4U_0^2 \frac{\sin^2 u}{u} \frac{2 \cos u}{2u}$$

$$(U_0 \frac{\sin u}{u})^2 (1 + 4 \cos^2 u + 4 \cos u \cos 8u)$$

$$\text{即光强 } I = I_0 (\frac{\sin u}{u})^2 [1 + 4 \cos^2 u + 4 \cos u \cos 8u]$$

九. 相对论动能

$$E_k = mc^2 - m_0 c^2$$

静止能量 $m_0 c^2$

$$mc^2 = 2m_0 c^2$$

$$\frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} c^2 = 2m_0 c^2 \quad v = \frac{\sqrt{3}}{2} c$$

$$\text{十. } W = 1.9 \text{ eV}$$

$$\nu = \frac{W}{h} = 4.59 \times 10^{14} \text{ Hz}$$

$$\lambda = \frac{c}{\nu} = 6.54 \times 10^{-7} \text{ m}$$

$$\frac{hc}{\lambda} - W = 2.4 \text{ eV} \quad \lambda = 2.99 \times 10^{-7} \text{ m}$$